APRIL 16TH, 2026

TEXAS
The University of Texas at Austin

THE ASYMMETRICAL IMPACT OF DAILY DISCRIMINATION ON MEXICAN-ORIGIN YOUTH'S EMOTIONAL AND PHYSICAL RESPONSES

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INTRODUCTION

BACKGROUNDS

- Mexican-origin youth is facing heightened discrimination in sociopolitical climates characterized by anti-immigrant sentiment (Herda, 2025).

- Discrimination may occur on a day-to-day basis (Williams et al., 2003) :

- Being treated with less courtesy
- Encountering negative stereotypes
- Being threatened by others on a specific day…

PRIOR RESEARCH

- **Prior studies used multilevel modeling approach to learn how daily discrimination influence youths' health:**
- Between-person (general discrimination exposure; person A vs. person B)
- Within-person (more/less discrimination on a day)

- **Higher levels of discrimination, on average or on a day, are associated with:**
- More negative emotion on average/on a specific day (Valentino et al., 2025)
- More physical symptoms on average/on a specific day (Wang & Yip, 2020)

RESEARCH GAP

1. Most studies describe the impact of discrimination as unidirectional:

- Emphasize intensified exposures on a specific day worsens youth outcomes
- Obscured how youth recover when discrimination decrease in another day.

Increased discrimination **undermine** well-being on a specific day

Add up to create stable between person difference

Decreased discrimination improve well-being on a specific day

"A relief day"
Help bounce back and recover

RESEARCH GAP

2. Traditional multilevel modeling approach hold symmetry assumption:

Increased discrimination **undermine** well-being on a specific day

Decreased discrimination improve well-being on a specific day

Negative bias perspective:

- *"bad is stronger than good"* (Baumeister et al., 2001)
- The magnitude of the entry bullying victimization > exit (Jang et al., 2024)

Theories of resilience in development (Masten, 2001):

- Positive fluctuations activate protective mechanisms, fostering recovery outweigh the impact of prior negative events (increased discrimination)

CURRENT STUDY

The asymmetrical effect of increase vs decrease of daily discrimination on Mexican-origin youth's daily emotional and physical health.

Hypothesis 1 Higher level of daily discrimination is associated with more negative emotion and physical symptoms at both within- and between- person level
Youth who experience more discrimination on average report worse health; On days when youth experience more discrimination than usual, they will report worse health.

Hypothesis 2 The increase and decrease of discrimination on a specific day (WP) influence youths' emotion and physical symptoms with different magnitude
Decreases in discrimination have stronger, weaker, or comparable effects relative to increases (exploratory analysis).

METHODS

Methods-Participants

- The data for this study come from a four-wave longitudinal study (Wave 1: 2012–2015, Wave 2: 2013–2016, Wave 3: 2017–2020, Wave 4: 2021–2024) examining Mexican immigrant families in central Texas

- At Wave 4, 245 youths participated in a **14-day daily diary study**:

1) A daily survey on personal devices
2) Daily experiences of discrimination, negative emotion, and physical symptoms

- Two were excluded due to incorrectly reporting the date on all 14 days and having response durations of less than three minutes 256 (88.5%) completed the survey on all four days

Methods-Measures

Daily Discrimination

- Four items adapted from the Daily Discrimination Scale (Kessler et al., 1999).
- Sample item: "I was treated with less courtesy than other people today."
- 1 (strongly disagree) to 5 (strongly agree). $\omega_{between} = 0.96$; $\omega_{within} = 0.76$

Negative Emotions

- McNair's Profile of Mood States Brief (POMS-B) (Shacham, 1983)
- Six negative emotions: nervous, on edge, uneasy, discouraged, sad, hopeless
- 1 (not at all) to 5 (very much), $\omega_{between} = 0.82$; $\omega_{within} = 0.97$

Physical Symptoms

- 6 physical symptoms: headache, cold/flu symptoms, fatigue, allergy symptoms, stomachache, other physical issues
- 0 (no) or 1 (yes). Summing responses across the six items

Covariates Age, Gender (0= male, 1= female), Nativity (0= Mexico-born, 1= US-born), Maternal education level: 1(Elementary and below) - 4(College/University +)

Methods-Analytic strategy

- **Person-mean centered** multilevel modeling was conducted in R using the lme4 package (Bates et al., 2003).

Hypothesis 1

Level 1 (Within-person level):

$$y_{ti} = \beta_{0i} + \beta_{1i}(x_{ti} - \bar{X}_i) + e_{ti}$$

Level 2 (Between-person level):

$$\beta_{0i} = \gamma_{00} + \gamma_{01}\bar{X}_i + (\gamma_{02,03,04\ldots}\, \textit{Person-level Covariates}) + U_{0i}$$ (random intercept)

$$\beta_{1i} = \gamma_{10}$$

Composite:

$$y_{ti} = \gamma_{00} + \gamma_{10}(x_{ti} - \bar{X}_i) + \gamma_{01}\bar{X}_i + \left(\gamma_{02,03\ldots}\, \textit{Person level Covariates}\right) + U_{0i} + e_{ti}$$

Methods-Analytic strategy Hypothesis 2

- **Asymmetry effects: coding procedure from developed by (Allison, 2019)**

family_id	day_of_study	tdis	dispos	disneg	tdiscumpos	tdiscumneg	discpos_btw	discpos_wth
10054	1	1.75	0.00	0.00	0.00	0.00	0.48076923	-0.48076923
10054	2	1.75	0.00	0.00	0.00	0.00	0.48076923	-0.48076923
10054	3	2.00	0.25	0.00	0.25	0.00	0.48076923	-0.23076923
10054	4	2.00	0.00	0.00	0.25	0.00	0.48076923	-0.23076923
10054	5	1.75	0.00	0.25	0.25	0.25	0.48076923	-0.23076923
10054	7	1.75	NA	NA	0.25	0.25	0.48076923	-0.23076923
10054	8	1.75	0.00	0.00	0.25	0.25	0.48076923	-0.23076923
10054	9	1.00	0.00	0.75	0.25	1.00	0.48076923	-0.23076923
10054	10	1.00	0.00	0.00	0.25	1.00	0.48076923	-0.23076923
10054	11	1.75	0.75	0.00	1.00	1.00	0.48076923	0.51923077
10054	12	1.75	0.00	0.00	1.00	1.00	0.48076923	0.51923077
10054	13	2.00	0.25	0.00	1.25	1.00	0.48076923	0.76923077
10054	14	1.00	0.00	1.00	1.25	2.00	0.48076923	0.76923077

Z_i and Z_i $Z_{it} - Z_i$ and $Z_{it} - Z_i$

Methods-Analytic strategy

- **Asymmetry effects**

Level 1 (Within-person level):

$$y_{it} = \beta_{0i} + \beta_{1i}^{+}\,(Z_{it}^{+} - \overline{Z_i^{+}}) + \beta_{1i}^{-}\,(Z_{it}^{-} - \overline{Z_i^{-}}) + e_{ti}$$

Level 2 (Between-person level):

$$\beta_{0i} = \gamma_{00} + \gamma_{01}^{+}\,\overline{Z_i^{+}} + \gamma_{01}^{-}\,\overline{Z_i^{-}} + (\gamma_{02,03\ldots}\,Person\text{-}level\ Covariates) + U_{0i}$$

$$\beta_{1i}^{+} = \gamma_{10}^{+}$$

$$\beta_{1i}^{-} = \gamma_{10}^{-}$$

Composite:

$$y_{ti} = \gamma_{00} + \boldsymbol{\gamma_{10}^{+}}\,(Z_{it}^{+} - \overline{Z_i^{+}}) + \boldsymbol{\gamma_{10}^{-}}(Z_{it}^{-} - \overline{Z_i^{-}}) + \gamma_{01}^{+}\,\overline{Z_i^{+}} + \gamma_{01}^{-}\,\overline{Z_i^{-}} + \left(\gamma_{02,03\ldots}\,Person - level\ Covariates\right) + U_{0i} + e_{ti}$$

Whether the effect of increased daily discrimination compared to yesterday is equal to the effect of decreased daily discrimination compared to yesterday.

Wald tests: $\gamma_{10}^{+} = -\gamma_{10}^{-}$

RESULTS

Descriptive statistics

ICC:
56% and 46.7% was attributable to between-person (BP) differences

44% and 53.3% was attributable to within-person (WP) variation

	Overall (N=243)
Adolescent_age	
Mean (SD)	20.9 (1.28)
Median [Min, Max]	20.9 [17.2, 25.0]
Missing	12 (4.9%)
Adolescent_gender	
Female	139 (57.2%)
Male	88 (36.2%)
Missing	16 (6.6%)
Adolescent_nativity	
non-U.S. born	59 (24.3%)
U.S. born	184 (75.7%)
Mother's education level	
No formal schooling	9 (3.7%)
Some elementary school	24 (9.9%)
Finished elementary school	38 (15.6%)
Some middle school/junior high school	27 (11.1%)
Finished middle school/junior high school	44 (18.1%)
Some high school	31 (12.8%)
Finished high school	26 (10.7%)
Finished technical or vocational training after high school	21 (8.6%)
Finished community college degree (AA)	5 (2.1%)
Finished university/bachelor degree (BA/BS)	13 (5.3%)
Finished graduate degree (Master degree, medical, etc.)	0 (0%)
Missing	5 (2.1%)

For negative feelings

Predictors	Discrimination to negative feelings			
	Model 1		Model 2	
	b (S.E.)	*p*	*b (S.E.)*	*p*
BP Discrimination	0.60 (0.07)	**<0.001**		
WP Discrimination	0.29 (0.02)	**<0.001**		
More Days with Increased Discrimination (BP)			0.15 (0.07)	**0.041**
More Days with Decreased Discrimination (BP)			0.05 (0.07)	0.449
Increased Discrimination than Yesterday (WP)[a]			0.18 (0.02)	**<0.001**
Decreased Discrimination than Yesterday (WP)[b]			-0.27 (0.02)	**<0.001**
Wald test for a = - b (asymmetry effect)			x^2=39.43	**<0.001**
Age	0.03 (0.03)	0.337	0.05 (0.03)	0.154
Born in Mexico	-0.06 (0.09)	0.458	-0.02 (0.10)	0.854
Female	0.15 (0.07)	**0.046**	0.13 (0.08)	0.112
(Intercept)	0.91 (0.60)	0.13	0.36 (0.68)	0.599
Random Effects				
Residual variance	0.29		0.29	
Random intercept variance	0.27		0.35	
N	226		226	
Observations	2643		2644	
Marginal R2/Conditional R2	0.199/0.586		0.081/0.585	
AIC	4778.985		4859.019	
log-Likelihood	-2381.492		-2419.51	

A significant asymmetry effect:

Decreases in discrimination were associated with a stronger reduction in negative feelings ($b = -0.27$, $SE = 0.02$, $p < .001$) than the increase in negative feelings associated with intensified discrimination ($b = 0.18$, $SE = 0.02$, $p < .001$)

For physical symptoms

Predictors	Discrimination to physical symptoms			
	Model 3		Model 4	
	b (S.E.)	*p*	*b (S.E.)*	*p*
BP Discrimination	0.47 (0.10)	**<0.001**		
WP Discrimination	0.11 (0.04)	**0.002**		
More Days with Increased Discrimination (BP)			0.13 (0.10)	0.203
More Days with Decreased Discrimination (BP)			-0.01 (0.10)	0.905
Increased Discrimination than Yesterday (WP)[a]			0.02 (0.04)	0.639
Decreased Discrimination than Yesterday (WP)[b]			-0.16 (0.04)	**<0.001**
Wald test for a = - b (asymmetry effect)			x^2=41.84	**<0.001**
Age	0.08 (0.04)	**0.055**	0.10 (0.05)	**0.03**
Born in Mexico	0.05 (0.13)	0.682	0.09 (0.14)	0.52
Female	0.43 (0.11)	**<0.001**	0.43 (0.12)	**<0.001**
(Intercept)	-0.67 (0.91)	0.466	-1.07 (0.95)	0.261
Random Effects				
Residual variance	0.76		0.75	
Random intercept variance	0.60		0.67	
N	226		226	
Observations	2643		2643	
Marginal R2/Conditional R2	0.087/0.490		0.058/0.501	
AIC	7337.164		7328.234	
log-Likelihood	-3660.582		-3654.117	

A significant asymmetry effect:

Decreases in discrimination were associated with significantly fewer physical symptoms ($b = -0.16$, $SE = 0.04$, $p < .001$).

Increases in discrimination compared to the previous day were not significantly associated with physical symptoms ($b = 0.02$, $SE = 0.04$, $p = .639$)

KEY TAKEAWAYS

- Higher BP and WP levels of daily discrimination were associated with more negative emotion and physical symptoms.

- Increase vs. decrease in daily discrimination have asymmetrical effects:
  Reductions in discrimination from one day to the next relate to larger improvements in emotional and physical symptoms than the worsening observed following increases.

When young people experience a day with more discrimination, they may feel worse, but a better day that follows can bring powerful relief, both emotionally and physically.

DISCUSSION

IMPLICATION & CONTRIBUTION

- **Adolescents' well-being fluctuates with contextual stress and moments of relief.**
- Daily discrimination fluctuated.
- Increases and decreases in discrimination have unequal impacts.

- **Short-term recovery processes is powerful and should be identified**
- Youth are not merely passive recipients of daily discrimination but exhibit active recovery responses once social environments become less hostile (Resilience).

- **Interventions might aim:**
- Not only preventing discriminatory incidents but enhancing recovery contexts
- Identify Mexican-origin youth who show blunted recovery patterns (an early indicator of stress dysregulation or burnout)

LIMITATIONS

- Sample consisted solely of Mexican-origin youths from Central Texas, which may limit the generalizability of the findings.

- **14-day design captures may not generalize to longer-term adaptation processes.**
- Do not know whether the asymmetry observed here accumulates over months or years, or whether the pattern reverses under chronic discrimination exposure.

- **Causality cannot be definitively established although use within-person design**
- Improvements in emotion/physical symptoms may influence how youth perceive or interpret discrimination.
- Time-lagged or reciprocal modelling may needed

THANKS

Advisor: Su Yeong Kim

Collaborators: Wen Wen & Haoxiang Yu

QUESTIONS